

A Simple Pharmacokinetic Model of Prenatal and Postnatal Exposure to Perfluoroalkyl Substances (PFASs)

Marc-André Verner,^{*,†,‡} Gérard Ngueta,[†] Elizabeth T. Jensen,^{§,⊥} Hermann Fromme,[#] Wolfgang Völkel,[#] Unni Cecilie Nygaard,[▽] Berit Granum,[▽] and Matthew P. Longnecker[§]

[†]Department of Occupational and Environmental Health, School of Public Health and [‡]Université de Montréal Public Health Research Institute (IRSPUM), Université de Montréal, Montreal, Quebec H3T 1A8, Canada

[§]National Institute of Environmental Health Sciences, National Institutes of Health, Department of Health and Human Services, Research Triangle Park, North Carolina 27709, United States

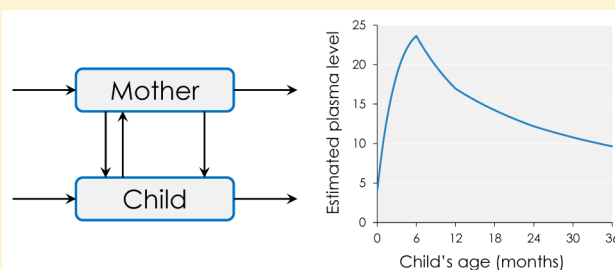
[⊥]Department of Epidemiology and Prevention, Division of Public Health Sciences, Wake Forest University School of Medicine, Winston-Salem, North Carolina 27157, United States

[#]Department of Chemical Safety and Toxicology, Bavarian Health and Food Safety Authority, D-80538 Munich, Germany

[▽]Division of Environmental Medicine, Norwegian Institute of Public Health, PO Box 4404 Nydalen, 0403 Oslo, Norway

S Supporting Information

ABSTRACT: Most children are exposed to perfluoroalkyl substances (PFASs) through placental transfer, breastfeeding, and other environmental sources. To date, there are no validated tools to estimate exposure and body burden during infancy and childhood. In this study, we aimed to (i) develop a two-generation pharmacokinetic model of prenatal and postnatal exposure to perfluorooctanoic acid (PFOA), perfluorooctanesulfonate (PFOS), and perfluorohexanesulfonate (PFHxS); and to (ii) evaluate it against measured children's levels in two studies. We developed a pharmacokinetic model consisting of a maternal and a child compartment to simulate lifetime exposure in women and transfer to the child across the placenta and through breastfeeding. To evaluate the model, we performed simulations for each mother–child dyad from two studies in which maternal PFAS levels at delivery and children's PFAS levels were available. Model predictions based on maternal PFAS levels, sex of child, body weight, and duration of breastfeeding explained between 52% and 60% of the variability in measured children's levels at 6 months of age and between 52% and 62% at 36 months. Monte Carlo simulations showed that the daily intake through breastfeeding and resulting internal PFAS levels can be much higher in nursing infants than in mothers. This pharmacokinetic model shows potential for postnatal exposure assessment in the context of epidemiological studies and risk assessment.



INTRODUCTION

Perfluoroalkyl substances (PFASs), such as perfluorooctanesulfonate (PFOS), perfluorooctanoic acid (PFOA), and perfluorohexanesulfonate (PFHxS), are synthetic fluorinated organic compounds used in industrial and consumer products due to their chemical and thermal stability and water and oil repellency.¹ Although the use of PFOS, PFOA, and PFHxS has been phased out in many countries, exposure to PFASs is ongoing, as is the assessment of risk due to exposure. The presence of PFASs in breast milk from women^{2–8} indicates that nursing infants are exposed during early development, a period during which they may be especially sensitive to chemicals.^{9,10} A recent epidemiological study of PFAS and immune function in children highlighted the importance of postnatal exposure: postnatal PFAS levels were more strongly associated with vaccine-specific antibody levels than prenatal PFAS levels.¹¹ Despite indications of the potential health effects of postnatal exposure to PFAS, most epidemiological studies focused on

prenatal exposure. This may reflect the challenge of exposure assessment during the postnatal period (i.e., blood sampling in infancy is impractical and there is a lack of tools to predict plasma levels).

Studies have shown that breastfeeding is an important route of PFAS excretion in breastfeeding mothers and, consequently, a source of exposure in nursing infants. In a study in which intake was estimated from PFAS levels in breast milk, dust, and air, Haug et al. (2011)¹² estimated that 94% and 83% of total PFOS and PFOA intake in 6 month olds is attributable to lactational exposure. Exposure through breastfeeding can lead to high internal exposure; in a study of 20 infants who were exclusively breastfed up to 6 months of age, plasma levels at 6

Received: September 9, 2015

Revised: December 15, 2015

Accepted: December 21, 2015

Published: December 21, 2015

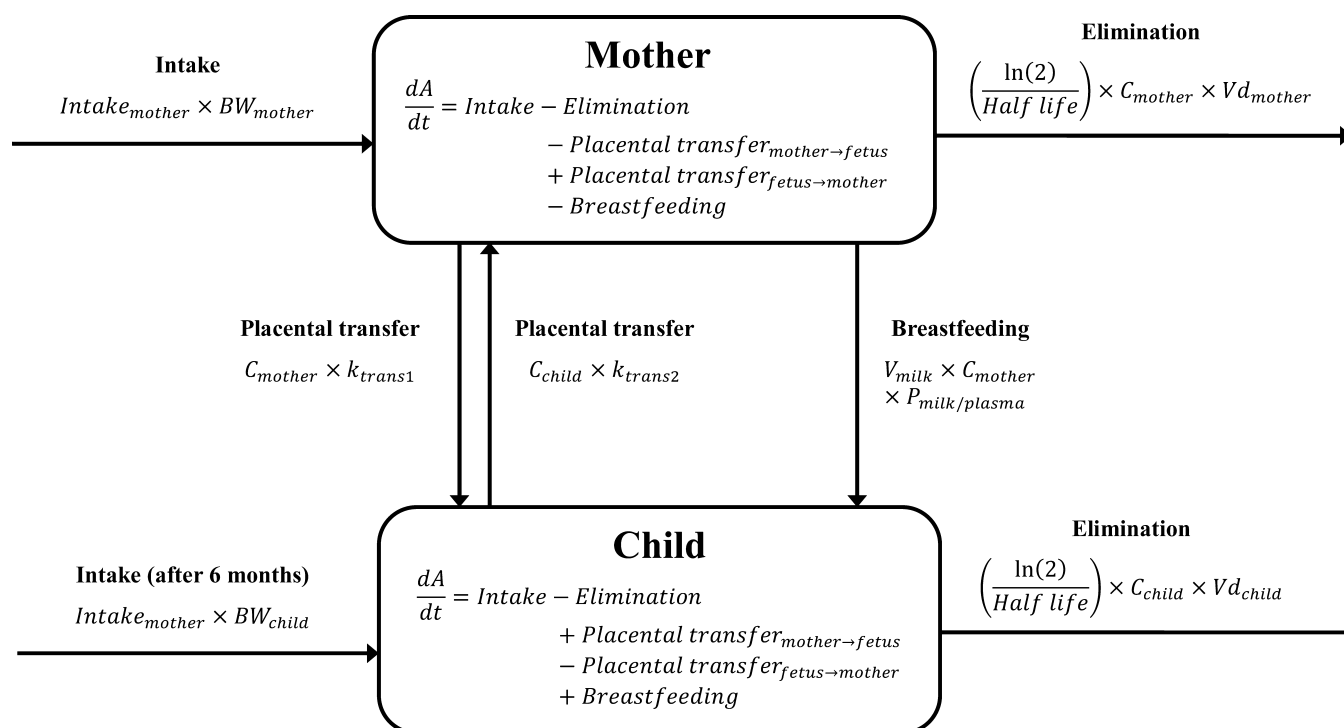


Figure 1. Conceptual representation and main equations of the pharmacokinetic model. Abbreviations: A = amount; BW = body weight; C = concentration; K_{trans} = placental diffusion rate; P = partition coefficient; V = volume; V_d = volume of distribution.

months were on average 4.0 times (PFOA) and 1.1 times (PFOS) higher than levels measured in maternal plasma at delivery.¹³ Studies also found a positive association between the duration of breastfeeding and children's PFAS levels,^{14,15} a finding that was paralleled by a decrease in maternal PFAS levels during breastfeeding.^{13,14,16,17} These results call for epidemiological studies on the potential health effects of lactational exposure to PFASs. In addition, this route of exposure will need to be accounted for during assessment of the risk of PFASs, namely in terms of relating maternal dose to child's dose. Tools to adequately estimate exposure and body burden in children will be instrumental to accomplish these tasks.

A physiologically based pharmacokinetic model for plasma concentration of PFOA and PFOS in infants and toddlers up to 15 months of age has been developed by Loccisano et al. (2013).¹⁸ For reasons of applicability in epidemiology and risk assessment, we decided to develop a novel pharmacokinetic model for which (1) prenatal and postnatal exposure can be accommodated within the same model (these two periods are covered by two separate models in Loccisano et al. (2013));¹⁸ (2) women's lifetime exposure is simulated (simulations using the Loccisano et al. (2013)¹⁸ models start at conception or delivery, and initial tissue concentrations need to be calculated using another PBPK model); (3) the maximum duration of breastfeeding is 30 months rather than 15 months; (4) levels in older children (3 years in this case) can be modeled; and (5) another compound, PFHxS, can be modeled.

Specifically, we developed a model to estimate women's exposure to three PFASs (PFOA, PFOS, and PFHxS), placental transfer during gestation, and postnatal exposure in children through breastfeeding and other sources. These three PFASs were selected because sufficient information on pharmacokinetic parameters was available in the literature. We also set out to evaluate the model by comparing predictions to plasma

levels measured in children. Finally, we present applications in epidemiology and risk assessment.

MATERIALS AND METHODS

Pharmacokinetic Model. We developed a simple pharmacokinetic model with two compartments: one for the mother and one for the child (Figure 1). We assumed PFASs to be fully absorbed through oral ingestion because exposure through other routes like dermal absorption and inhalation are thought to be negligible in comparison.¹⁹ We used volumes of distribution of 170 mL/kg of body weight for PFOA and 230 mL/kg of body weight for PFOS as estimated by Thompson et al. (2010).²⁰ These values were similar to those calculated from controlled exposures in monkeys.^{21,22} Because the volume of distribution for PFHxS has not been estimated for humans, we used a value of 213 mL/kg of body weight derived from a monkey study.²³ Body-weight profiles for the mother and child were based on median growth curves from the Center for Disease Control and Prevention (CDC).²⁴ These growth curves were adjusted to fit individual-specific measured weight, e.g., for a woman with a prepregnancy body weight of 65 kg at 20 years of age (versus the median weight at that age, 54 kg), the median growth curve was shifted by a factor of 1.2. PFAS elimination from maternal and child compartments was calculated based on published arithmetic mean half-lives of 3.8 years for PFOA, 5.4 years for PFOS, and 8.5 years for PFHxS.²⁵ We assumed partition of PFAS between fetal and maternal plasma during pregnancy to reach an equilibrium rapidly, with a resulting concentration ratio equal to the average ratio from studies reviewed by Aylward et al. (2014).²⁶ We calculated mother–fetus (k_{trans1}) and fetus–mother (k_{trans2}) transfer rates (L/h) so that their ratio ($k_{\text{trans1}}/k_{\text{trans2}}$) is the same as measured ratios; k_{trans1} was given the same value as the reported maternal/cord plasma level ratio (0.78 for PFOA, 0.45

Table 1. Characteristics of Participants in the German ($n = 27$) and Norwegian studies ($n = 51$) Included for Pharmacokinetic Model Evaluation

parameter	German study			Norwegian study		
	N (%)	mean $\pm$ SD	range	N (%)	mean $\pm$ SD	range
maternal age at delivery (years)		34 $\pm$ 4	28–42		32 $\pm$ 4	25–42
maternal prepregnancy weight (kg)		59 $\pm$ 6	49–75		66 $\pm$ 9	50–97
sex of child						
female	–			23 (45)		
male	–			28 (55)		
child's weight at 36 months (kg)		–	–		15.3 $\pm$ 1.6	12–20
duration of breastfeeding (months)		6	–		13.2 $\pm$ 4.8	0–28
maternal PFOA level at delivery (ng/mL)		2.4 $\pm$ 1.4	0.6–7.0		1.1 $\pm$ 0.4	0.4–2.1
child PFOA levels at 6 months (ng/mL)		8.7 $\pm$ 5.6	1.0–26.9		–	–
child PFOA levels at 19 months (ng/mL)		5.7 $\pm$ 2.5	2.9–11.9		–	–
child PFOA levels at 36 months (ng/mL)		–	–		2.8 $\pm$ 1.0	1.0–5.9
maternal PFOS levels at delivery (ng/mL)		3.5 $\pm$ 1.7	0.8–9.4		5.4 $\pm$ 2.0	1.4–10.7
child PFOS levels at 6 months (ng/mL)		3.6 $\pm$ 2.1	0.7–9.6		–	–
child PFOS levels at 19 months (ng/mL)		2.4 $\pm$ 1.5	1.0–4.8		–	–
child PFOS levels at 36 months (ng/mL)		–	–		5.0 $\pm$ 2.0	1.4–9.6
maternal PFHxS levels at delivery (ng/mL)		0.6 $\pm$ 0.3	0.2–1.4		0.4 $\pm$ 0.5	0.1–2.8
child PFHxS levels at 6 months (ng/mL)		0.8 $\pm$ 0.4	0.2–1.8		–	–
child PFHxS levels at 19 months (ng/mL)		0.7 $\pm$ 0.2	0.5–1.2		–	–
child PFHxS levels at 36 months (ng/mL)		–	–		0.8 $\pm$ 1.0	0.2–6.8

for PFOS, and 0.56 for PFHxS), and $k_{\text{trans}2}$ was given a value of 1.00. Transfer to and from the fetus was calculated as the product between the plasma concentration and the transfer rate (Figure 1). We modeled exposure through lactation based on the duration of breastfeeding, breast milk consumption rates, and milk/plasma ratios. Breast milk intake was based on the equation derived by Arcus-Arth et al.(2005)²⁷ for the first 12 months of life [eq 1]:

$$\text{intake (g/kg/d)} = -0.312 \times \text{age (days)} + 157.7 \quad (1)$$

The equation to describe breast milk intake from 12 to 30 months was derived through linear regression of intake estimates from three studies^{28–30} [eq 2]:

$$\text{intake(g/d)} = -0.763 \times \text{age (days)} + 720.3 \quad (2)$$

Units for eq 1 (ng/kg/day) and eq 2 (ng/day) are different because data on daily milk intake past 12 months were not reported on a body-weight basis. Milk/plasma ratios ($P_{\text{milk/plasma}}$) were calculated by averaging ratios from studies in which plasma and milk samples were taken at the same time or within a few days. The ratios were 0.058 for PFOA,^{12,31,32} 0.014 for PFOS,^{12,31–33} and 0.014 for PFHxS.^{31,33} Transfer through breastfeeding was defined as the product of the volume of milk, the PFAS level in the maternal compartment, and the $P_{\text{milk/plasma}}$ (Figure 1). Given the lack of data on exposure to PFAS in infancy and childhood, we assumed daily intake (ng/kg/day) from sources other than breastfeeding to be equal to that of the mother after 6 months of age.

We developed the model so it can generate profiles of children's PFAS levels based on a few model inputs:

- maternal age at delivery (years);
- maternal prepregnancy body weight (kg);
- sex of child (because growth patterns are sex-specific);
- child's birth weight (kg);
- child's weight (kg) at any given time point (years) during childhood; and

- total duration of breastfeeding (regardless of exclusivity) (years).

We conducted a global sensitivity analysis to identify sensitive model parameters and inputs (see Figure S1 for methods and results). The pharmacokinetic model was coded using acslX (Aegis Technologies Inc., Huntsville, AL). The model code is available in the Supporting Information.

Data for Model Evaluation. We evaluated the model by comparing estimated children's PFAS levels to measured children's levels from two studies. In the first longitudinal biomonitoring study, which was based in Germany, blood was sampled from mothers (pregnancy [$n = 44$], delivery [$n = 38$], 6 months postpartum [$n = 47$]) and children (delivery [$n = 33$], 6 months [$n = 40$], and 19 months [$n = 24$]).¹³ However, information on breastfeeding in this study was limited to whether children were exclusively, predominantly, partially, or not breastfed at 6 months of age. The second study was a substudy of the Norwegian Mother and Child Cohort Study³⁴ on prenatal PFAS exposure and children's immune response to vaccination and immune-related health effects. In this study, blood samples were taken from mothers at delivery ($n = 99$) and from children at 36 months of age ($n = 51$).³⁵ Information on the total duration of breastfeeding was collected when the child was 3 years old, which is when a blood specimen was obtained from the child. Both studies measured multiple PFASs, including PFOA, PFOS, and PFHxS.

Model Evaluation. We generated profiles of PFAS levels for children enrolled in the German and Norwegian studies. For each mother–child dyad, we used individual-specific model inputs (see Table 1) to simulate exposure. We first calibrated a maternal daily intake using the pharmacokinetic model so that simulated PFAS levels in the maternal compartment at the time of delivery matched measured levels. Then, using this estimated maternal daily intake, we used the model to estimate children's PFAS levels across the first three years of life. Model precision was evaluated using the coefficient of determination (R^2) from linear regression models of measured children's PFAS levels versus estimated levels (regression models were allowed to have

an intercept other than 0). Precision was defined as the proportion of variability in measured PFAS levels explained by model predictions. Because information on the duration of breastfeeding was limited in the German study, we only included children who were exclusively, predominantly, or partially breastfed until 6 months (i.e., children who were not breastfed at 6 months were excluded, leaving 27 subjects in the analysis). Information on breastfeeding after 6 months was not available in this study; therefore, levels at 19 months were not used to evaluate model precision. We performed sensitivity analyses to evaluate model precision for exposure scenarios in which children's daily intake (ng/kg/day) was 0.5, 1.0 (main analyses), or 2.0 times the maternal daily intake after 6 months of age.

Child/Mother Ratios of PFAS Levels and Daily Intake.

The relation between children's PFAS levels and daily intake to those of mothers is important to characterize when a risk assessment is conducted. We performed Monte Carlo simulations for each compound to estimate the variability in child/mother ratios of PFAS plasma levels and daily intake. At each of the 10 000 iterations, model inputs and parameters were sampled from probabilistic distributions (see Table 2). This set of values was subsequently used to simulate PFAS levels in the mother and child for a maternal daily intake of 1

ng/kg/day. Simulations were performed assuming a duration of breastfeeding of 30 months to evaluate maximum lactational exposure in children. For the child/mother PFAS level ratios, we calculated the distribution of child PFAS levels throughout the first 36 months and maternal levels at delivery. Profiles of child/mother-level ratios were compared to ratios calculated from levels measured in the German and Norwegian studies at 6, 19, and 36 months. Child/mother daily intake (ng/kg/day) ratios were calculated based on simultaneous daily intake in mothers and children.

RESULTS

Global sensitivity analyses indicated that parameters related to breastfeeding (i.e., duration of breastfeeding, volume of milk, and milk/plasma partition) were the most sensitive, followed by the volume of distribution, cord/mother partition, half-life, and maternal age at delivery (see Table S1). To evaluate model precision, we generated profiles of exposure to PFASs in mothers and children based on maternal levels at delivery and compared children's estimated levels to levels measured at 6 months of age (German study) and at 36 months of age (Norwegian study). Mean ($\pm$ SD) daily doses in mothers (ng/day) estimated with the pharmacokinetic model to match measured plasma levels at delivery were 13.0 ($\pm$ 7.6) for PFOA, 18.1 ($\pm$ 9.0) for PFOS, and 2.1 ($\pm$ 0.9) for PFHxS in the German study and 6.7 ($\pm$ 3.1) for PFOA, 31.7 ($\pm$ 12.1) for PFOS, and 1.4 ($\pm$ 1.9) for PFHxS in the Norwegian study.

The precision of the PFAS concentrations estimated by the model varied across compounds and children's age. At 6 months of age in the German study, levels estimated with the model explained 60% of the variability in PFOA levels, 53% of the variability in PFOS levels, and 57% of the variability in PFHxS levels (Figure 2). On average, model predictions slightly overestimated PFOA and PFOS levels measured at 6 months (Table S1), an overestimation that was most evident in a small subset of the participants (Figure 2). At 36 months of age in the Norwegian study, estimated levels explained 62% of the variability in PFOA levels, 52% of the variability in PFOS levels, and 53% of the variability in PFHxS levels (because PFHxS levels were skewed, they were log₁₀-transformed prior to R² calculation) (Figure 3). When we excluded an outlying measured PFHxS level at 36 months (6 SDs from the mean), the model precision increased to 68%. On average, there was a slight underestimation of measured PFOA levels and a clear underestimation of PFHxS at 36 months (Table S1); on average, measured levels at 36 months were 2.8 times higher than estimated levels (excluding the outlier, which was 20 times higher than the estimated level). This underestimation was not observed at 6 months. Model precision was essentially the same for the different scenarios of children's daily intake after 6 months of age (Table S2). However, estimated PFOA levels at 36 months of age were no longer underestimated when we assumed the children's daily intake (ng/kg/day) to be twice their mother's.

Monte Carlo simulations indicated that children's PFAS levels can be higher than levels measured in mothers (Figure 4). For example, PFOA levels in children peaked during the first year, with child/mother ratios reaching 4.5 at the 50th percentile, 7.8 at the 95th percentile, and a maximum of 15.3. Child/mother-level ratios were lower for PFOS and PFHxS, which have milk/plasma ratios (0.014) about four times lower than that of PFOA (0.058). When comparing results from Monte Carlo simulations to data from the German and

Table 2. Parameter Distributions for the Monte Carlo Simulations

parameter	distribution	mean $\pm$ SD
maternal age at delivery (years)	normal	25 ^a $\pm$ 6 ^a
maternal prepregnancy body weight (kg)	normal	75 ^b $\pm$ 18 ^b
sex of child	Bernoulli ($p = 0.5$)	
child's weight (multiplier ^c)	normal	1.0 $\pm$ 0.12 ^d
PFOA $P_{\text{milk/plasma}}$	log normal	0.058 ^e $\pm$ 0.035 ^f
PFOS $P_{\text{milk/plasma}}$	log normal	0.014 ^g $\pm$ 0.009 ^f
PFHxS $P_{\text{milk/plasma}}$	log normal	0.014 ^h $\pm$ 0.006 ^f
PFOA Volume of distribution (mL/kg)	normal	170 ⁱ $\pm$ 26 ^j
PFOS Volume of distribution (mL/kg)	normal	230 ⁱ $\pm$ 35 ^j
PFHxS Volume of distribution (mL/kg)	normal	213 ^k $\pm$ 32 ^j
PFOA Half-life (years)	log normal	3.8 ^l $\pm$ 1.7 ^l
PFOS Half-life (years)	log normal	5.5 ^l $\pm$ 3.4 ^l
PFHxS Half-life (years)	log normal	8.4 ^l $\pm$ 5.2 ^l
PFOA $P_{\text{cord/mother}}$	normal	0.78 ^m $\pm$ 0.15 ⁿ
PFOS $P_{\text{cord/mother}}$	normal	0.45 ^m $\pm$ 0.13 ⁿ
PFHxS $P_{\text{cord/mother}}$	normal	0.56 ^m $\pm$ 0.11 ⁿ
volume of breast milk (multiplier ^c)	normal	1.00 $\pm$ 0.17 ^o

^aMartin et al. (2012)⁴⁷ ^bStandard deviation calculated from the interquartile range (SD = 0.74 $\times$ IQR).⁴⁸ ^cParameters that fluctuate within individual simulations were varied using a multiplier with a distribution centered around 1.0, a standard deviation equal to the coefficient of variation. ^dSD calculated from data in Kuczmarski et al. (2000)²⁴ ^eMean calculated from three studies.^{12,31,32} ^fSD calculated from individual data in Kim et al. (2011)³¹ ^gMean calculated from four studies.^{12,31–33} ^hMean calculated from two studies.^{31,33} ⁱTaken from Thompson et al. (2010)²⁰ ^jBecause no data was available on the variability and uncertainty in the volume of distribution, we assumed a coefficient of variation of 15%. ^kFrom a study in monkeys.²³ ^lTaken from Olsen et al. (2007)²⁵ ^mAverage of studies reviewed by Aylward et al. (2014)²⁶ ⁿSD calculated from individual data in Kim et al. (2011)³¹ ^oSD calculated using data from Arcus-Arth et al. (2005)²⁷

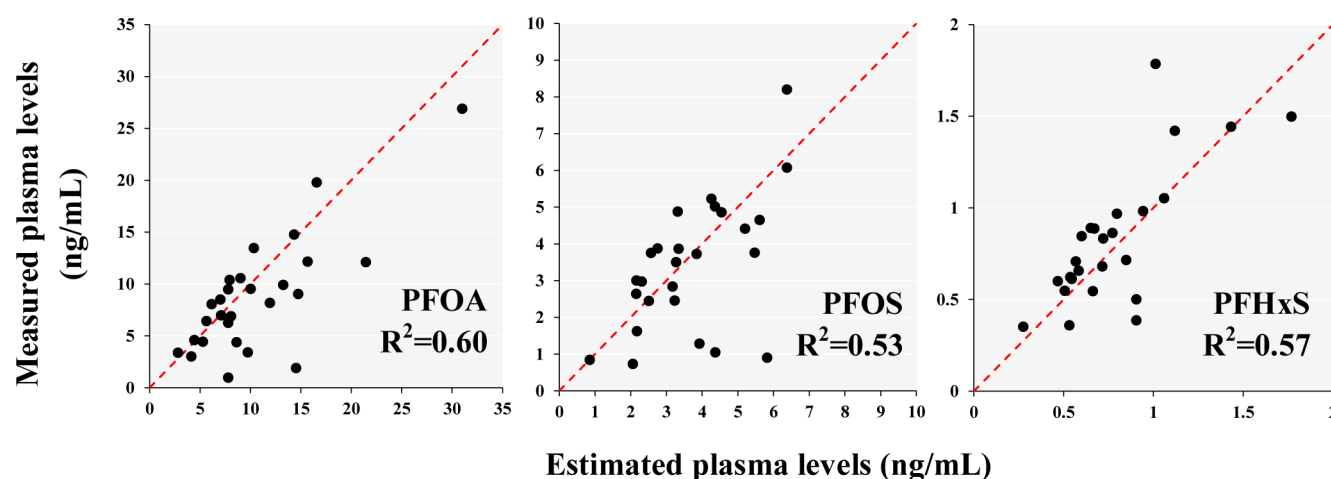


Figure 2. Measured plasma PFAS levels in 6 month olds from the Fromme et al. (2010)¹ study versus levels estimated using the pharmacokinetic model.

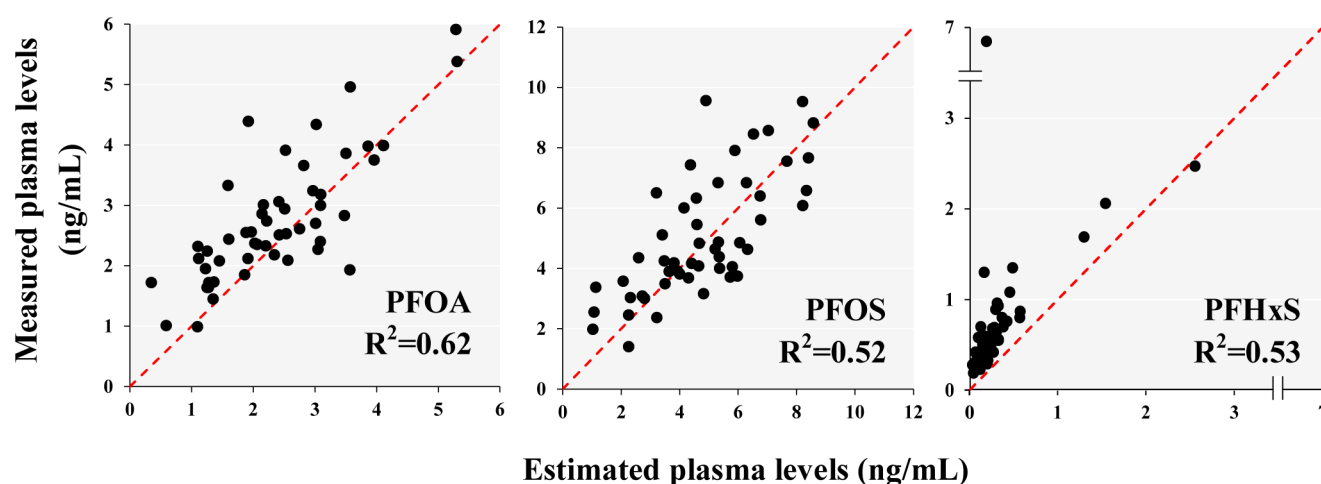


Figure 3. Measured plasma PFAS levels in 3 year olds from the Granum et al. (2013)¹ study versus levels estimated using the pharmacokinetic model. Note: Because of the skewed distribution of PFHxS levels, the R^2 was calculated on log-transformed data.

Norwegian studies, we found that most measured child/mother level ratios fell below the 95th percentile of simulated ratios, with the exception of PFHxS at 36 months when an important proportion (51%) of measured child/mother level ratios were above the 95th percentile of simulated ratios. Peak child/mother dose (ng/kg/day) ratios were observed right after birth and declined thereafter. The 95th percentile of child/mother dose ratios reached maximum values of 9.6 for PFOA, 2.4 for PFOS, and 2.6 for PFHxS.

DISCUSSION

In this study, we elaborated a simple two-compartment pharmacokinetic model of prenatal and postnatal exposure to PFASs. This model allows the reconstruction of children's PFAS levels through the first three years of life in the context of epidemiological studies and risk assessment.

We generated PFAS level profiles for each of the mother-child dyads participating in two studies based on maternal plasma PFAS levels at the time of delivery, information on physiology and duration of breastfeeding. The daily intake for each mother was calibrated using the pharmacokinetic model to obtain matching predicted and measured plasma PFAS levels at delivery. PFAS daily intakes estimated in German women were

much lower than those estimated in a German duplicate diet study³⁶ for PFOA (13 versus 220 ng/day) and PFOS (18 versus 112 ng/day). The corresponding plasma concentrations in the two studies showed similar differences for PFOA (4.6 versus 2.0 ng/mL) and PFOS (13.5 versus 3.0 ng/mL). The much greater ratio of estimated intakes of PFOA may be partially due to the half-life we used (3.8 years); others have suggested a lower value (e.g., 2.3 years³⁷). Estimated daily intakes in the Norwegian women were closer to daily intakes estimated in a Norwegian study of dietary exposure;³⁸ we estimated daily intakes of 6.7 ng/day for PFOA, 31.7 ng/day for PFOS, and 1.4 ng/day for PFHxS, whereas intakes estimated in the dietary exposure study were 31.0 ng/day for PFOA, 18.0 ng/day for PFOS, and 1.2 ng/day for PFHxS.

Levels estimated with the pharmacokinetic model explained between 52% and 60% of the variability in measured levels at 6 months in the German study and between 52% and 62% of the variability in measured levels at 36 months in the Norwegian study. Highest model precision was observed for PFOA, which partitions more readily into breast milk than PFOS and PFHxS. Inability to predict a larger proportion of the variability in children's levels could be the result of our modeling approach, uncertainty in model parameters, and variability in children's exposure through sources other than breastfeeding. Our simple

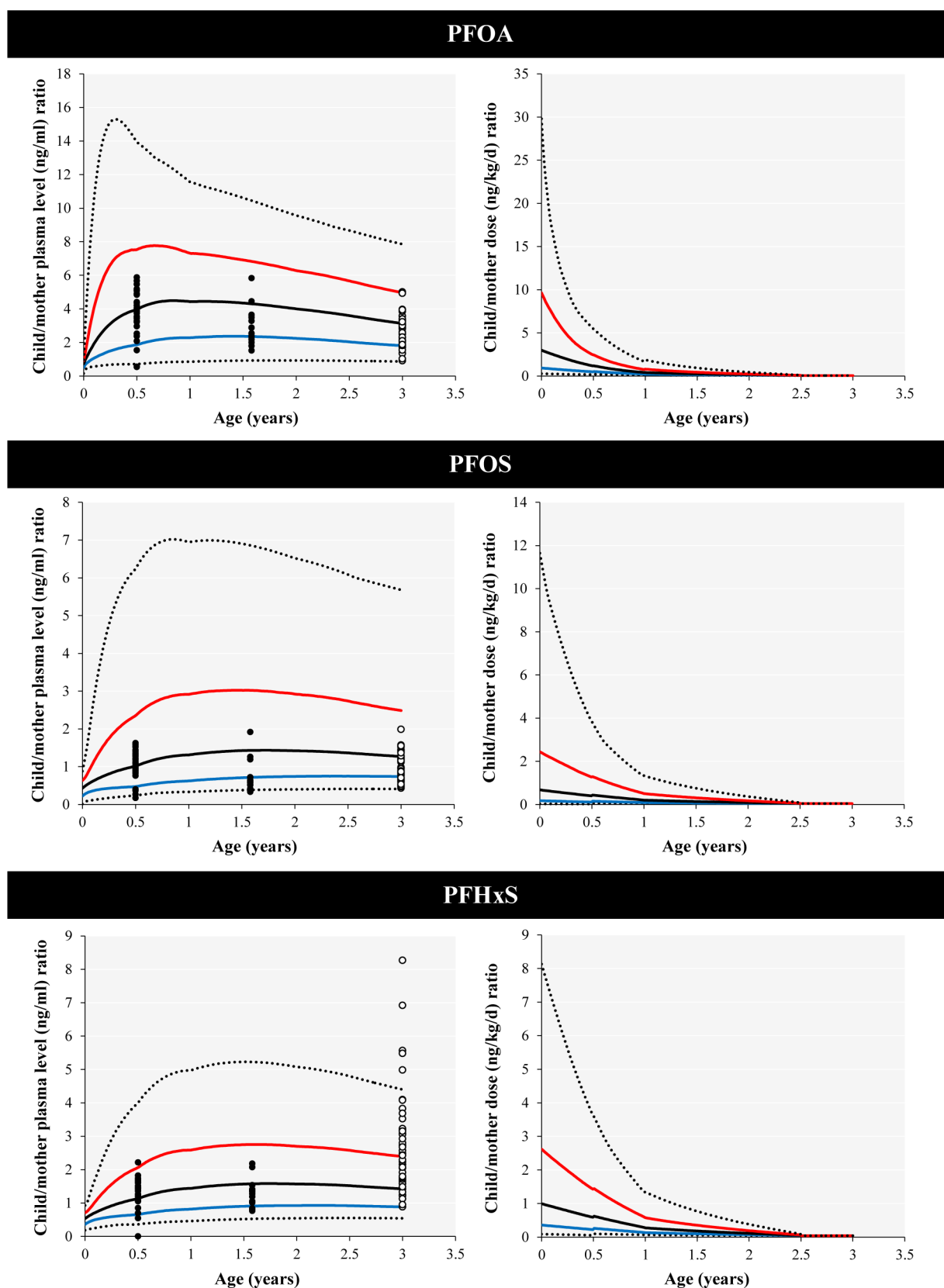


Figure 4. Monte Carlo simulations ($n = 10\,000$) of child/mother ratios of PFAS levels (ng/mL) and daily intake (ng/kg/day) for a breastfeeding period of 30 months. The black line represent the 50th percentile, the blue line represents the 5th percentile, the red line represents the 95th percentile, and the dotted lines represent minimum and maximum values. Dots represent ratios of measured maternal and child PFAS levels from the German study (●) and the Norwegian study (○). Note: one outlier for PFHxS (child/mother level ratio of 22) was excluded from this figure.

pharmacokinetic model does not account for many physiological processes associated with pregnancy and lactation like the increase in blood volume during pregnancy. Although similar precision was observed for simple and complex physiologically based pharmacokinetic (PBPK) models of lipophilic persistent organic pollutants previously,^{39,40} we cannot rule out the possibility that a life-stage PBPK model of PFAS would provide more precise estimates. Partitioning of PFASs into breast milk was parametrized as the average of values from published studies; the reported values varied substantially from one study to the other, e.g., for PFOA, the milk/plasma partition varied from 0.038³² to 0.11.³² In addition, we did not account for variations in the partition of PFASs between plasma and milk across months of breastfeeding as breast milk composition changes (e.g., lipids and proteins).⁴¹ Also, our model did not account for isomer-specific (e.g., linear, branched) pharmacokinetics because (i) we could not find data on the transfer into breast milk for the different isomers, and (ii) the two studies used for model evaluation did not report isomer-specific concentrations. Different isomers of PFAS are known to have different binding to albumin,⁴² urinary clearance,⁴³ and transplacental transfer.⁴⁴ Should data on milk/plasma partition of the different PFAS isomers become available, it could be included to refine the model. Another limitation is that we did not have sufficient information to reliably distinguish exclusive breastfeeding and mixed feeding. PFAS intake through food, water, and dust may also contribute significantly to children's levels. Our assumption that children's daily intake (ng/kg/day) other than through breastfeeding was equal to that of mother may be an oversimplification. Where additional information that might improve the estimate of children's exposure is available (e.g., food frequency questionnaires, and dust measurements), inclusion of this information in the model could lead to improved prediction of PFAS concentrations. The underestimation of PFHxS levels at 36 months suggests that sources other than breast milk may be especially important for this compound. Finally, the volumes of distribution used in our model (for mothers and children) were scaled according to body weight. This simplification may have reduced model precision because it does not account for changes in body composition across development.

Our Monte Carlo simulations showed that PFAS levels in children can exceed maternal levels during and after lactation. Measured child/mother level ratios mostly fell below the 95th percentile for the first two years of life, suggesting that the 95th percentile of simulated levels is a reasonable estimation of maximum exposure between birth and 24 months in the context of risk assessment. However, levels of PFHxS at 36 months of age in the Norwegian study were higher than levels predicted by the model. These results, in conjunction with the fact that the model could predict PFHxS child/mother level ratios accurately at 6 and 19 months, suggest that environmental exposure to this compound may substantially contribute to children's levels after the second year of life.

A pair of main applications are foreseen with this pharmacokinetic model: estimation of postnatal exposure to PFOA, PFOS, and PFHxS in longitudinal birth cohort studies and risk assessment. Collecting blood samples in infancy and childhood is often challenging or impractical in epidemiological studies. Our model allows generating complete time-courses of PFAS levels in children from which different exposure metrics can be derived (e.g., the monthly PFAS level, the area under the curve for a given period, and the maximum concentration)

depending on the outcome of interest and the hypothesized time- and concentration-dependent association (Figure 5).

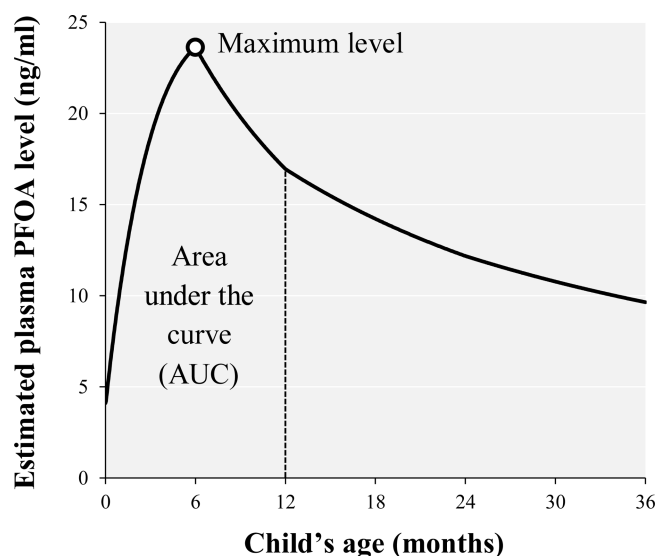


Figure 5. Exposure metrics that can be derived from the time-course of estimated PFOA levels to evaluate exposure–outcome associations in epidemiological studies. In this example, metrics include the plasma PFOA level at any time point, the maximum level, and the area under the curve (AUC) for the 0–12 month period.

Although model predictions do not fully explain variability in true children's plasma levels ($R^2 = 52\text{--}62$), a recent study of polychlorinated biphenyls and immune function suggests that this level of model precision may be sufficient to detect exposure associations. In that study, a pharmacokinetic model explaining 59% of the variability in measured plasma PCB-153 at 6 months in a Slovak cohort enabled the identification of a time-dependent association between PCB-153 levels at 6 months and a decrease in BCG-specific antibody levels.⁴⁵ In terms of risk assessment, understanding the relationship between maternal exposure and levels and children's exposure and levels could help adjust reference doses to protect children. A similar approach was recently employed to relate the maternal daily dose of hexachlorobenzene to infants' exposure through breastfeeding.⁴⁶ Of note is the fact that the model does not account for exposure to PFAS precursors (e.g., perfluorooctane sulfonamido ethanols), which may be biotransformed to PFAS in the body.

In conclusion, our pharmacokinetic model allows the estimation of children's exposure and body burden of PFOA, PFOS, and PFHxS throughout the first three years with a precision ranging from 52%–62%. Future uses include assessing children's PFAS levels in longitudinal birth cohort studies in which collecting biological specimens is impractical and relating maternal dose to child's dose and body burden for risk assessment. Whether the model precision is sufficient to detect associations in epidemiological studies will depend on the resolution of input parameters (e.g., the duration of breastfeeding) and the strength of the associations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.5b04399.

A global sensitivity analysis of the pharmacokinetic model, comparison between measured and estimated PFAS levels, model precision and accuracy for different scenarios of children's PFAS intake, and model code. (PDF)

AUTHOR INFORMATION

Corresponding Author

*Tel: 514-343-6465; fax: 514-343-2200; e-mail: marc-andre.verner.1@umontreal.ca.

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research was supported in part by the Intramural Research Program of the NIH, National Institute of Environmental Health Sciences. M.A.V. is the recipient of an Emerging Researcher Fellowship from the Université de Montréal Public Health Research Institute (IRSPUM). The Norwegian Mother and Child Cohort Study is supported by the Norwegian Ministry of Health and the Ministry of Education and Research, the National Institutes of Health (NIEHS) (contract no. NO-ES-75558), the National Institutes of Health (NINDS) (grant no. 1 UO1 NS 047537-01), and the Norwegian Research Council/FUGE (grant no. 151918/S10).

REFERENCES

- (1) Buck, R. C.; Franklin, J.; Berger, U.; Conder, J. M.; Cousins, I. T.; de Voogt, P.; Jensen, A. A.; Kannan, K.; Mabury, S. A.; van Leeuwen, S. P. Perfluoroalkyl and polyfluoroalkyl substances in the environment: terminology, classification, and origins. *Integr. Environ. Assess. Manage.* **2011**, *7* (4), 513–41.
- (2) Croes, K.; Colles, A.; Koppen, G.; Govarts, E.; Bruckers, L.; Van de Mieroop, E.; Nelen, V.; Covaci, A.; Dirtu, A. C.; Thomsen, C.; Haug, L. S.; Becher, G.; Mampaey, M.; Schoeters, G.; Van Larebeke, N.; Baeyens, W. Persistent organic pollutants (POPs) in human milk: a biomonitoring study in rural areas of Flanders (Belgium). *Chemosphere* **2012**, *89* (8), 988–94.
- (3) Barbarossa, A.; Masetti, R.; Gazzotti, T.; Zama, D.; Astolfi, A.; Veyrand, B.; Pession, A.; Pagliuca, G. Perfluoroalkyl substances in human milk: a first survey in Italy. *Environ. Int.* **2013**, *51*, 27–30.
- (4) Volkel, W.; Genzel-Boroviczeny, O.; Demmelmair, H.; Gebauer, C.; Koletzko, B.; Twardella, D.; Raab, U.; Fromme, H. Perfluorooctane sulphonate (PFOS) and perfluorooctanoic acid (PFOA) in human breast milk: results of a pilot study. *Int. J. Hyg. Environ. Health* **2008**, *211* (3–4), 440–6.
- (5) Fujii, Y.; Yan, J.; Harada, K. H.; Hitomi, T.; Yang, H.; Wang, P.; Koizumi, A. Levels and profiles of long-chain perfluorinated carboxylic acids in human breast milk and infant formulas in East Asia. *Chemosphere* **2012**, *86* (3), 315–21.
- (6) Sundstrom, M.; Ehresman, D. J.; Bignert, A.; Butenhoff, J. L.; Olsen, G. W.; Chang, S. C.; Bergman, A. A temporal trend study (1972–2008) of perfluorooctanesulfonate, perfluorohexanesulfonate, and perfluorooctanoate in pooled human milk samples from Stockholm, Sweden. *Environ. Int.* **2011**, *37* (1), 178–83.
- (7) Tao, L.; Kannan, K.; Wong, C. M.; Arcaro, K. F.; Butenhoff, J. L. Perfluorinated compounds in human milk from Massachusetts, U.S.A. *Environ. Sci. Technol.* **2008**, *42* (8), 3096–101.
- (8) Raab, U.; Albrecht, M.; Preiss, U.; Volkel, W.; Schwegler, U.; Fromme, H. Organochlorine compounds, nitro musks and perfluorinated substances in breast milk - results from Bavarian Monitoring of Breast Milk 2007/8. *Chemosphere* **2013**, *93* (3), 461–7.
- (9) Rice, D.; Barone, S., Jr. Critical periods of vulnerability for the developing nervous system: evidence from humans and animal models. *Environ. Health Perspect* **2000**, *108* (3), 511–33.
- (10) Dietert, R. R.; Etzel, R. A.; Chen, D.; Halonen, M.; Holladay, S. D.; Jarabek, A. M.; Landreth, K.; Peden, D. B.; Pinkerton, K.; Smialowicz, R. J.; Zoetis, T. Workshop to identify critical windows of exposure for children's health: immune and respiratory systems work group summary. *Environ. Health Perspect* **2000**, *108* (3), 483–90.
- (11) Grandjean, P.; Andersen, E. W.; Budtz-Jorgensen, E.; Nielsen, F.; Molbak, K.; Weihe, P.; Heilmann, C. Serum vaccine antibody concentrations in children exposed to perfluorinated compounds. *Jama* **2012**, *307* (4), 391–7.
- (12) Haug, L. S.; Huber, S.; Becher, G.; Thomsen, C. Characterisation of human exposure pathways to perfluorinated compounds—comparing exposure estimates with biomarkers of exposure. *Environ. Int.* **2011**, *37* (4), 687–93.
- (13) Fromme, H.; Mosch, C.; Morovitz, M.; Alba-Alejandre, I.; Boehmer, S.; Kiranoglu, M.; Faber, F.; Hannibal, I.; Genzel-Boroviczeny, O.; Koletzko, B.; Volkel, W. Pre- and postnatal exposure to perfluorinated compounds (PFCs). *Environ. Sci. Technol.* **2010**, *44* (18), 7123–9.
- (14) Mondal, D.; Weldon, R. H.; Armstrong, B. G.; Gibson, L. J.; Lopez-Espinosa, M. J.; Shin, H. M.; Fletcher, T. Breastfeeding: a potential excretion route for mothers and implications for infant exposure to perfluoroalkyl acids. *Environ. Health Perspect* **2014**, *122* (2), 187–92.
- (15) Pinney, S. M.; Biro, F. M.; Windham, G. C.; Herrick, R. L.; Yaghjian, L.; Calafat, A. M.; Succop, P.; Sucharew, H.; Ball, K. M.; Kato, K.; Kushi, L. H.; Bornschein, R. Serum biomarkers of polyfluoroalkyl compound exposure in young girls in Greater Cincinnati and the San Francisco Bay Area, USA. *Environ. Pollut.* **2014**, *184*, 327–34.
- (16) Jain, R. B. Effect of pregnancy on the levels of selected perfluoroalkyl compounds for females aged 17–39 years: data from National Health and Nutrition Examination Survey 2003–2008. *J. Toxicol. Environ. Health, Part A* **2013**, *76* (7), 409–21.
- (17) von Ehrenstein, O. S.; Fenton, S. E.; Kato, K.; Kuklenyik, Z.; Calafat, A. M.; Hines, E. P. Polyfluoroalkyl chemicals in the serum and milk of breastfeeding women. *Reprod. Toxicol.* **2009**, *27* (3–4), 239–45.
- (18) Loccisano, A. E.; Longnecker, M. P.; Campbell, J. L., Jr.; Andersen, M. E.; Clewell, H. J., 3rd Development of PBPK models for PFOA and PFOS for human pregnancy and lactation life stages. *J. Toxicol. Environ. Health, Part A* **2013**, *76* (1), 25–57.
- (19) Lorber, M.; Egeghy, P. P. Simple intake and pharmacokinetic modeling to characterize exposure of Americans to perfluorooctanoic acid, PFOA. *Environ. Sci. Technol.* **2011**, *45* (19), 8006–14.
- (20) Thompson, J.; Lorber, M.; Toms, L. M.; Kato, K.; Calafat, A. M.; Mueller, J. F. Use of simple pharmacokinetic modeling to characterize exposure of Australians to perfluorooctanoic acid and perfluorooctane sulfonic acid. *Environ. Int.* **2010**, *36* (4), 390–7.
- (21) Andersen, M. E.; Clewell, H. J., 3rd; Tan, Y. M.; Butenhoff, J. L.; Olsen, G. W. Pharmacokinetic modeling of saturable, renal resorption of perfluoroalkylacids in monkeys—probing the determinants of long plasma half-lives. *Toxicology* **2006**, *227* (1–2), 156–64.
- (22) Butenhoff, J. L.; Kennedy, G. L., Jr.; Hinderliter, P. M.; Lieder, P. H.; Jung, R.; Hansen, K. J.; Gorman, G. S.; Noker, P. E.; Thomford, P. J. Pharmacokinetics of perfluorooctanoate in cynomolgus monkeys. *Toxicol. Sci.* **2004**, *82* (2), 394–406.
- (23) Sundstrom, M.; Chang, S. C.; Noker, P. E.; Gorman, G. S.; Hart, J. A.; Ehresman, D. J.; Bergman, A.; Butenhoff, J. L. Comparative pharmacokinetics of perfluorohexanesulfonate (PFHxS) in rats, mice, and monkeys. *Reprod. Toxicol.* **2012**, *33* (4), 441–51.
- (24) Kuczmarowski, R. J.; Ogden, C. L.; Grummer-Strawn, L. M.; Flegal, K. M.; Guo, S. S.; Wei, R.; Mei, Z.; Curtin, L. R.; Roche, A. F.; Johnson, C. L. CDC growth charts: United States. *Adv. Data* **2000**, *314*, 1–27.

- (25) Olsen, G. W.; Burris, J. M.; Ehresman, D. J.; Froehlich, J. W.; Seacat, A. M.; Butenhoff, J. L.; Zobel, L. R. Half-life of serum elimination of perfluorooctanesulfonate, perfluorohexanesulfonate, and perfluorooctanoate in retired fluorochemical production workers. *Environ. Health Perspect.* **2007**, *115* (9), 1298–305.
- (26) Aylward, L. L.; Hays, S. M.; Kirman, C. R.; Marchitti, S. A.; Kenneke, J. F.; English, C.; Mattison, D. R.; Becker, R. A. Relationships of chemical concentrations in maternal and cord blood: a review of available data. *J. Toxicol. Environ. Health, Part B* **2014**, *17* (3), 175–203.
- (27) Arcus-Arth, A.; Krowech, G.; Zeise, L. Breast milk and lipid intake distributions for assessing cumulative exposure and risk. *J. Exposure Anal. Environ. Epidemiol.* **2005**, *15* (4), 357–65.
- (28) Dewey, K. G.; Finley, D. A.; Lonnerdal, B. Breast milk volume and composition during late lactation (7–20 months). *J. Pediatr. Gastroenterol. Nutr.* **1984**, *3* (5), 713–20.
- (29) Hennart, P.; Delogne-Desnoeck, J.; Vis, H.; Robyn, C. Serum levels of prolactin and milk production in women during a lactation period of thirty months. *Clin. Endocrinol.* **1981**, *14* (4), 349–53.
- (30) Kent, J. C.; Mitoulas, L.; Cox, D. B.; Owens, R. A.; Hartmann, P. E. Breast volume and milk production during extended lactation in women. *Experimental physiology* **1999**, *84* (2), 435–47.
- (31) Kim, S. K.; Lee, K. T.; Kang, C. S.; Tao, L.; Kannan, K.; Kim, K. R.; Kim, C. K.; Lee, J. S.; Park, P. S.; Yoo, Y. W.; Ha, J. Y.; Shin, Y. S.; Lee, J. H. Distribution of perfluorochemicals between sera and milk from the same mothers and implications for prenatal and postnatal exposures. *Environ. Pollut.* **2011**, *159* (1), 169–74.
- (32) Liu, J.; Li, J.; Liu, Y.; Chan, H. M.; Zhao, Y.; Cai, Z.; Wu, Y. Comparison on gestation and lactation exposure of perfluorinated compounds for newborns. *Environ. Int.* **2011**, *37* (7), 1206–12.
- (33) Karrman, A.; Ericson, I.; van Bavel, B.; Darnerud, P. O.; Aune, M.; Glynn, A.; Lignell, S.; Lindstrom, G. Exposure of perfluorinated chemicals through lactation: levels of matched human milk and serum and a temporal trend, 1996–2004, in Sweden. *Environ. Health Perspect.* **2007**, *115* (2), 226–230.
- (34) Magnus, P.; Irgens, L. M.; Haug, K.; Nystad, W.; Skjaerven, R.; Stoltenberg, C. MoBa Study, G., Cohort profile: the Norwegian Mother and Child Cohort Study (MoBa). *International journal of epidemiology* **2006**, *35* (5), 1146–50.
- (35) Granum, B.; Haug, L. S.; Namork, E.; Stolevik, S. B.; Thomsen, C.; Aaberge, I. S.; van Loveren, H.; Lovik, M.; Nygaard, U. C. Pre-natal exposure to perfluoroalkyl substances may be associated with altered vaccine antibody levels and immune-related health outcomes in early childhood. *J. Immunotoxicol.* **2013**, *10* (4), 373–9.
- (36) Fromme, H.; Schlummer, M.; Moller, A.; Gruber, L.; Wolz, G.; Ungewiss, J.; Bohmer, S.; Dekant, W.; Mayer, R.; Liebl, B.; Twardella, D. Exposure of an adult population to perfluorinated substances using duplicate diet portions and biomonitoring data. *Environ. Sci. Technol.* **2007**, *41* (22), 7928–7933.
- (37) Bartell, S. M.; Calafat, A. M.; Lyu, C.; Kato, K.; Ryan, P. B.; Steenland, K. Rate of decline in serum PFOA concentrations after granular activated carbon filtration at two public water systems in Ohio and West Virginia. *Environ. Health Perspect.* **2010**, *118* (2), 222–8.
- (38) Haug, L. S.; Salihovic, S.; Jogsten, I. E.; Thomsen, C.; van Bavel, B.; Lindstrom, G.; Becher, G. Levels in food and beverages and daily intake of perfluorinated compounds in Norway. *Chemosphere* **2010**, *80* (10), 1137–43.
- (39) Verner, M. A.; Ayotte, P.; Muckle, G.; Charbonneau, M.; Haddad, S. A physiologically based pharmacokinetic model for the assessment of infant exposure to persistent organic pollutants in epidemiologic studies. *Environ. Health Perspect.* **2009**, *117* (3), 481–7.
- (40) Verner, M. A.; Sonneborn, D.; Lancz, K.; Muckle, G.; Ayotte, P.; Dewailly, E.; Kocan, A.; Palkovicova, L.; Trnovec, T.; Haddad, S.; Hertz-Picciotto, I.; Eggesbo, M. Toxicokinetic modeling of persistent organic pollutant levels in blood from birth to 45 months of age in longitudinal birth cohort studies. *Environ. Health Perspect.* **2013**, *121* (1), 131–7.
- (41) Neville, M. C.; Allen, J. C.; Archer, P. C.; Casey, C. E.; Seacat, J.; Keller, R. P.; Lutes, V.; Rasbach, J.; Neifert, M. Studies in human lactation: milk volume and nutrient composition during weaning and lactogenesis. *Am. J. Clin. Nutr.* **1991**, *54* (1), 81–92.
- (42) Beesoon, S.; Martin, J. W. Isomer-Specific Binding Affinity of Perfluorooctanesulfonate (PFOS) and Perfluorooctanoate (PFOA) to Serum Proteins. *Environ. Sci. Technol.* **2015**, *49* (9), 5722–31.
- (43) Gao, Y.; Fu, J.; Cao, H.; Wang, Y.; Zhang, A.; Liang, Y.; Wang, T.; Zhao, C.; Jiang, G. Differential accumulation and elimination behavior of perfluoroalkyl Acid isomers in occupational workers in a manufactory in China. *Environ. Sci. Technol.* **2015**, *49* (11), 6953–62.
- (44) Beesoon, S.; Webster, G. M.; Shoeib, M.; Harner, T.; Benskin, J. P.; Martin, J. W. Isomer profiles of perfluorochemicals in matched maternal, cord, and house dust samples: manufacturing sources and transplacental transfer. *Environ. Health Perspect.* **2011**, *119* (11), 1659–64.
- (45) Jusko, T. A.; De Roos, A. J.; Lee, S. Y.; Thevenet-Morrison, K.; Schwartz, S. M.; Verner, M. A.; Palkovicova Murinova, L.; Drobna, B.; Kocan, A.; Fabisikova, A.; Conka, K.; Trnovec, T.; Hertz-Picciotto, I.; Lawrence, B. P. A Birth Cohort Study of Maternal and Infant Serum PCB-153 and DDE Concentrations and Responses to Infant Tuberculosis Vaccination. *Environ. Health Perspect.* **2015**, DOI: 10.1289/ehp.1510101.
- (46) Lehmann, G. M.; Verner, M. A.; Luukinen, B.; Henning, C.; Assimon, S. A.; LaKind, J. S.; McLanahan, E. D.; Phillips, L. J.; Davis, M. H.; Powers, C. M.; Hines, E. P.; Haddad, S.; Longnecker, M. P.; Poulsen, M. T.; Farrer, D. G.; Marchitti, S. A.; Tan, Y. M.; Swartout, J. C.; Sagiv, S. K.; Welsh, C.; Campbell, J. L., Jr.; Foster, W. G.; Yang, R. S.; Fenton, S. E.; Tornero-Velez, R.; Francis, B. M.; Barnett, J. B.; El-Masri, H. A.; Simmons, J. E. Improving the risk assessment of lipophilic persistent environmental chemicals in breast milk. *Crit. Rev. Toxicol.* **2014**, *44* (7), 600–17.
- (47) Martin, J.; Hamilton, B.; Ventura, S.; Osterdam, M.; Wilson, E.; Mathews, T. *Births: Final Data for 2010*; National Center for Health Statistics: Atlanta, GA; 2012.
- (48) McDowell, M. A.; Fryar, C. D.; Ogden, C. L.; Flegal, K. M. Anthropometric reference data for children and adults: United States, 2003–2006. *National health statistics reports* **2008**, *10*, 1–48.